

Twenty-metre optical paths in sub-19 cm rings: chord-skip toroidal multipass cells built from catalogue mirrors for drone-borne methane sensing

Draft v5 for internal review (simulation results) — 2 July 2026 — not for distribution

Sumant Varanasi¹, [co-authors — confirm], and Thomas Benoy^{2,*}

¹Indian Institute of Technology Delhi – Abu Dhabi, UAE ²Aston Institute of Photonic Technologies, Aston University, UK *t.benoy@aston.ac.uk

Abstract

Toroidal multipass cells fold long absorption paths into compact, mechanically robust volumes, but published designs demonstrate at most ~ 10 m of optical path in the 100–200 mm diameter class and typically require custom diamond-turned optics. We present a family of ring-geometry multipass cells that reach a verified 20.4–24.8 m optical path length within a 180 mm assembly envelope — and 20.6 m within 159 mm — using 13–16 catalogue 25.4 mm concave mirrors ($\approx$ US\$1000 of optics) and a sub-kilogram optical head. Three design principles enable this: (i) a chord-skip ring geometry whose near-diameter chords deliver up to 143 mm of path per reflection at $7\text{--}13^\circ$ incidence; (ii) an engineered dual-plane re-entrance condition in which the machined ring radius acts as the closure-tuning parameter (≈ 1.2 rad of accumulated transverse phase per millimetre), with a number-theoretic constellation rule that guarantees fringe-safe spot separation on every mirror; and (iii) mode-matched injection through a single 1.3 mm coupling hole, which makes hole losses negligible so that cell transmission equals $R^{(n-1)}$ exactly in mirror reflectivity R — 86.7 % for the headline 20.4 m design at $R = 0.999$. Designs are verified with an exact three-dimensional ray tracer including astigmatic Gaussian-beam propagation, cross-validated against an independent ray tracer to sub-micrometre agreement, and toleranced by Monte-Carlo analysis: at research-grade build tolerances the headline design completes all 144 traversals in 100 % of trials, the ± 1 % catalogue tolerance on mirror focal length is absorbed by a linear 0.72 mm-per-percent trim of the ring radius, and a plain aluminium ring holds alignment over ± 26 K. Because the mirrors provide $R = 0.999$, the photon budget admits ~ 700 reflections and the true limit is spot-pattern geometry under build error; a tolerance-tiered menu results, with every tier verified by per-trial physical criteria (path completion, spot merging, hole leakage, unassisted exit): 20.4 m at 86.7 % is robust at standard laboratory build tolerances; 29.0 m at 76.6 % ($\varnothing 183$ mm), 24.8 m at 83.9 %, and 20.7 m at 81.6 % in a $\varnothing 141$ mm envelope are robust at flight-grade (0.1 mrad) build tolerances — the glued/welded construction class a vibrating airframe requires regardless; and the geometric ceiling — 38.6 m in $\varnothing 169$ mm at 72.5 % — is reachable only with active alignment. Demanding build-walk-sized clearances during the search itself measurably hardens the surviving designs (the headline design's worst-case spot separation grows 30 %), turning tolerance from a post-hoc filter into a design input.

1. Introduction

Tunable diode laser absorption spectroscopy (TDLAS) of the methane $2\nu_3$ band near 1654 nm requires tens of metres of absorption path for sub-ppm sensitivity, while drone deployment restricts the sensor head to roughly a 19 cm envelope and about a kilogram of payload. Multipass cells reconcile these demands by folding the path between mirrors. The classical two-mirror Herriott cell [11–13] achieves this with re-entrant elliptical patterns; its densest variants reach 54.6 m in a 17 cm astigmatic cell [9] and 26.4 m in a

12 cm dense-pattern cell [10], at the cost of custom mirror pairs and critical two-mirror alignment. Ring-geometry (toroidal/circular) cells trade some packing density for one-piece mechanical robustness: demonstrated paths are 2.2–4.1 m in the monolithic toroidal cell of Tuzson et al. [1, 2], >12 m in a 145 mm circular paraboloid [3], 10 m at <200 g in the segmented cell of Graf et al. [4], and 10 m (30 m theoretical) from multi-layer patterns on a 100 mm ring [7]; mask-based fringe suppression [5], double-row geometries at larger diameter [6], and cryogenic toroidal cavities [8] extend the family. Across this literature, demonstrated toroidal-class paths in drone-compatible envelopes cluster at or below ~10 m, and every design relies on custom optics.

This work asks a deliberately constrained question: how much verified optical path can a ring cell deliver inside a 190 mm envelope using only catalogue mirrors — Thorlabs CM254-series 25.4 mm concave mirrors (16 focal lengths, ≈US\$74 each) — with 8–16 mirrors, a single 1.3 mm coupling hole, and a 1.3 mm collimated input? The chord-skip parameter is the entry point: in our early design-space exploration, moving from adjacent-mirror propagation to a skip of $s = 4$ on a 9-mirror ring raised the path length by 67 % while cutting the incidence angle from 67.5° to 10° , and a companion catalogue study showed 12.7 mm mirrors saturate near 12 m — fixing the present 25.4 mm, $N = 8$ –16 design space. Our contributions are: (1) the chord-skip parametrisation itself, with $s \approx N/2$ giving near-diameter chords at near-normal incidence; (2) an engineered — rather than searched-for — re-entrance condition, tuned by the machined ring radius against catalogue-locked curvatures, with an exact constellation criterion guaranteeing fringe-safe spot patterns before any ray is traced; (3) mode-matched single-hole coupling that renders the hole lossless; and (4) a fully verified design menu (13.6 m in 143 mm to 24.8 m in 180 mm), each design passing an eight-item physical check matrix in exact ray tracing, cross-validated independently, Monte-Carlo tolerated, and accompanied by machining-level build specifications and a drone mass budget.

2. Cell architecture and design principles

2.1 Chord-skip ring geometry

N identical concave mirrors of curvature radius R and clear-aperture radius 11.4 mm face inward from a ring of radius R_{ring} , one every $360^\circ/N$. A beam injected through a hole in mirror M_0 advances by s mirrors per reflection (the chord skip, $\gcd(N, s) = 1$). Each chord has length $L = 2 R_{\text{ring}} \sin(\pi s/N)$ and the incidence angle on every mirror is $\theta_i = \pi/2 - \pi s/N$. For $s \approx N/2$ the chords approach the ring diameter while θ_i falls to 6–13°: a 72 mm-radius ring yields 140+ mm of path per reflection — the geometric core of the result. The assembly envelope is $2(R_{\text{ring}} + 18 \text{ mm})$, the allowance covering substrate and housing wall.

2.2 Re-entrance: dual-plane phase closure

In the paraxial unit cell (one chord plus one mirror) the transverse ray oscillation advances by a phase θ per bounce, with $\cos \theta_t = 1 - L/(R \cos \theta_i)$ tangentially and $\cos \theta_s = 1 - L \cos \theta_i / R$ sagittally (the off-axis astigmatic split). The beam exits through the entrance hole after $n = kN$ reflections when both accumulated phases return to their launch value; zero-crossing launches close at phase 0 or π , giving four closure-mode combinations per geometry. The single machinable parameter R_{ring} moves both accumulated phases along a nearly fixed direction (≈ 1.2 rad per millimetre here), so a candidate is closable only if its signed phase residuals lie near that line; the residual is absorbed by the launch amplitudes through their weak aberration-mediated phase pull. Re-entrance thus becomes an engineered property: the ring is machined to the radius that closes the pattern for the delivered mirror lot, and the verified designs re-cross the hole centre to within 10^{-6} – 10^{-4} mm.

2.3 Constellation rule: fringe-safe spot patterns

At closure the $n = kN$ spots sample a Lissajous curve at unit-fraction phase steps. Three exact geometric rules follow: k must be odd (for even k the tangential π -slot coincides with a sagittal near-return and a spot lands in the coupling hole); $\gcd(M, k) = 1$ in both planes with $M = \text{round}(n\theta/2\pi)$ (otherwise the k spots per mirror collapse into clusters); and — decisive in practice — every mirror carries the same k -point pattern at a different phase origin along the curve, so the worst spot pair must be evaluated over all N mirrors, not only the launch mirror, which overestimates usable separation by up to $3\times$. The minimum pair distance and pattern extent are computed exactly per candidate, sizing the required amplitude and rejecting self-crowding patterns before ray tracing. Spot-overlap fringes — the dominant practical noise source in this cell class [5, 7] — are thereby excluded by construction: every design keeps every spot pair on every mirror at least the sum of the local $1/e^2$ radii plus 0.36–0.42 mm apart.

2.4 Mode-matched single-hole coupling

A 1.3 mm waist placed at the hole is badly mismatched to the cell eigenmode (mirror-plane radius $w = \sqrt{M^2\lambda L/\pi \sin \theta} \approx 0.3\text{--}0.45$ mm here); the beam then breathes to $\sim 3\times$ its injected size, spots overlap, and the hole clips 13.5 % of the power at entry and again at exit. Instead, the injection optics place the in-cell waist (0.20–0.33 mm) near mid-chord: the beam rides the eigenmode at 0.21–0.42 mm for the entire path, the radius at the hole is $\sim 0.3\text{--}0.37$ mm, and the 1.3 mm hole transmits $\approx 100\%$ both ways. Cell transmission is then exactly $T(R) = R^{(n-1)}$. At closure the round-trip ABCD map is the identity, so the exit beam reproduces the injected Gaussian and leaves through the same hole, separated from the input axis by $2\theta_i$ (22.5° for $N = 16$), placing laser and detector side by side behind M0 with no beamsplitter.

2.5 Injection chain: source and collimator selection

The mode-matched launch is a finite-conjugate imaging task — relay the source mode to a 0.20–0.33 mm waist roughly 45–120 mm past the coupling hole — and the collimator choice follows from it. The baseline chain is a fibre-pigtailed 1654 nm DFB (e.g. an Eblana single-mode device): the fibre mode (radius $\approx 5.5\ \mu\text{m}$) is imaged by a short-focal aspheric lens of the standard collimation-package type [18] operated slightly defocused, at magnification $\approx 60\times$. For the headline design an $f \approx 2$ mm asphere places the 0.33 mm waist ≈ 120 mm from the lens — the exact conjugates, computed by the q-parameter design, are listed on each specification sheet — so a single catalogue lens in an adjustable FC/APC collimator body performs collimation and mode matching in one element, and its focus adjustment is the fine trim used during alignment. An attractive alternative first stage is a reflective (off-axis-paraboloid) fibre collimator [17]: being achromatic and figure-stable it removes the lens focal drift from the thermal budget of §5.2, complementing the ± 26 K window of the aluminium ring; it is followed by a weak mode-matching lens. Direct free-space collimation of a TO-can device [18] is the most compact option but is quantitatively limited: the best residual divergence of a collimated TO60 package is $\approx 0.8^\circ$ full angle (≈ 7 mrad half-angle), whereas the cell eigenmode accepts $\lambda/(\pi w_0) \approx 1.6\text{--}2.6$ mrad — a $\approx 4\times$ mismatch, equivalent to an M^2 well beyond the verified 1.3 budget, costing $\geq 80\%$ of the light and destroying the spot-size guarantees. TO-can sources are therefore suitable only with additional spatial filtering (or for much shorter cells); the fibre-coupled variant is the design baseline.

3. Numerical methods

Designs were generated and verified with our open simulation platform: an exact 3-D ray tracer on toroidal/spherical surfaces (analytic normals, Newton-iterated intersections, per-bounce aperture tests), astigmatic Gaussian propagation via per-plane complex-q ABCD transfer with the actual per-bounce incidence angles and an M^2 factor, loss budget, per-plane stability, and spot diagnostics; earlier iterations of the platform added Sobol design-space sampling and machine-learning surrogates used in the exploratory phase. The search runs in two stages: Stage A enumerates the catalogue space ($16 \text{ curvatures} \times N = 8\text{--}16 \times \text{coprime skips} \times \text{a fine ring-radius grid} \times \text{odd } k$) against packing, envelope, stability, closure-residual and constellation criteria; Stage B ray-traces survivors, sweeps the ring radius in $20 \text{ }\mu\text{m}$ steps to locate the $\sim 20 \text{ }\mu\text{m}$ -wide closure valley, and polishes six launch/geometry parameters by Nelder-Mead. A design is accepted only if the final trace passes all eight checks of Table 2. The tracer was cross-validated against the independent open-source ray tracer Optiland [15]: over 64 bounces of the headline design the two agree to $0.000 \text{ }\mu\text{m}$ RMS in spot position, 0.01° in incidence angle, and $0.8 \text{ }\mu\text{m}$ in chord length.

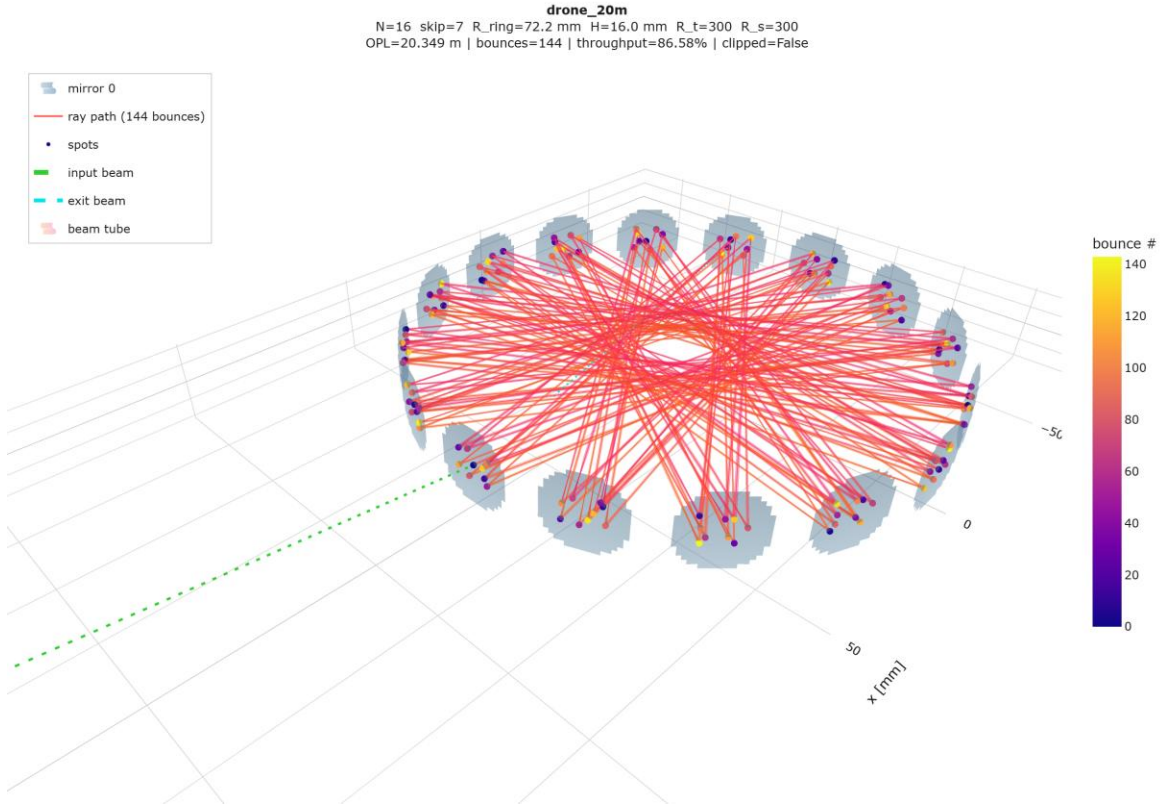


Fig. 1. Exact traced beam path of the headline design: 16 catalogue mirrors on a 72.155 mm ring, 144 chords (20.38 m), coloured by bounce order; the input beam (green) enters through the 1.3 mm hole in M0 and the closed pattern exits through the same hole.

4. Results

4.1 Verified design menu

Table 1. Verified design menu. Every row passes the full check matrix in the exact ray trace. Transmission $T = R^{(\text{chords}-1)}$ exactly (hole losses ≈ 0); quoted at $R = 0.999$ with sensitivity to coating in brackets ($R = 0.985 / 0.97$). Envelope = $2(R_{\text{ring}} + 18 \text{ mm})$.

design	mirrors	chords	OPL [m]	T @0.999	env. × height [mm]	robust at
geometric ceiling	14 × CM254-100 (R 200)	322	38.60	72.5 %	169 × 24	active align.
flight max OPL	12 × CM254-750 (R 1500)	204	28.99	76.6 %	183 × 23	flight build
long	16 × CM254-200 (R 400)	176	24.77	83.9 %	180 × 22	flight build
mid	14 × CM254-100 (R 200)	182	22.25	83.4 %	172 × 26	active align.
flight compact	12 × CM254-500 (R 1000)	204	20.66	81.6 %	141 × 27	flight build
headline 20 m	16 × CM254-150 (R 300)	144	20.38	86.7 %	180 × 16	standard build
balanced	13 × CM254-150 (R 300)	143	16.60	86.8 %	160 × 20	flight build
small	10 × CM254-150 (R 300)	190	14.85	82.8 %	133 × 19	flight build
small max-T	12 × CM254-250 (R 500)	132	13.64	87.7 %	143 × 16	flight build

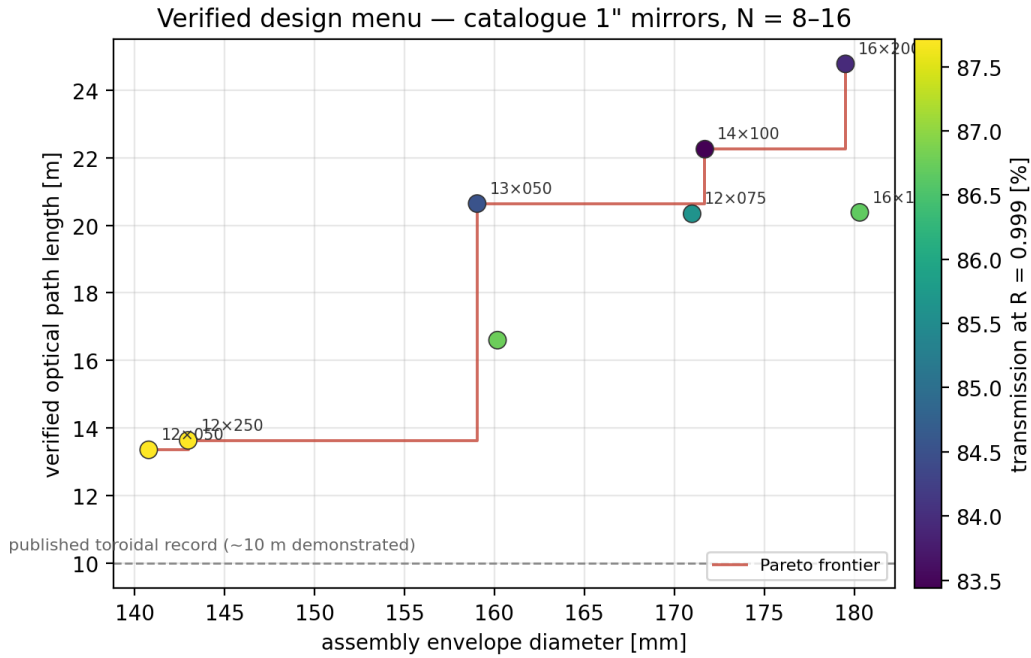


Fig. 2. The verified design menu as an envelope–path map (marker colour = transmission at R = 0.999). The Pareto frontier is populated entirely by catalogue-mirror designs; the dashed line marks the ~10 m demonstrated record of published toroidal cells in this size class.

The mirrors used in this project provide $R = 0.999$ at 1654 nm, so the photon budget admits ~700 reflections before transmission falls below 50 % and the real limit becomes spot-pattern geometry under build error. Table 1 therefore reports each design's ROBUSTNESS TIER, determined by a 100-trial Monte-Carlo at the stated build grade with per-trial physical criteria (full path completion, no spot merging by the simulator's own minimum-separation metric, positive hole clearance, and an exit that still leaves the 1.3 mm hole with no realignment): 'standard build' = kinematic-mount laboratory tolerances (0.5 mrad tilt class); 'flight build' = glued/welded precision tolerances (0.1 mrad class — the vibration spec a drone cell needs anyway); 'active alignment' = nominally feasible but requiring in-situ trim (ring temperature + launch adjuster). (Transmission for any other coating follows exactly from $T(R) = R^{(\text{chords}-1)}$, Fig. 7 — e.g. a protected-gold build at $R = 0.985$ would transmit 6–14 %.) The tier boundaries, established by exhaustive search (0.1 mm ring-radius grid, up to 720 bounces, $k \leq 45$ spots per mirror, with clearances sized to the expected

build-error spot walk), are themselves the key observation. At standard tolerances the dense long-path constellations lose spot identity — relative spot motion of 0.3–0.5 mm collapses their separation margins (the 38.6 m design's 5th-percentile minimum separation falls to 0.06 mm) — while at flight grade the 176–204-chord designs hold every margin with $\geq 99\%$ path completion. Because pattern walk grows roughly with the square root of the chord count, the standard-tolerance ceiling sits near 145 chords on the 11.4 mm clear aperture. Downward, the smallest robust envelope is 133 mm (14.9 m); below ~ 113 mm, nine 25.4 mm mirrors no longer pack. The architecture's map is therefore 20.4 m per standard build, 29.0 m per precision build, and 38.6 m per active alignment; larger-aperture mirrors are the identified route past all three.

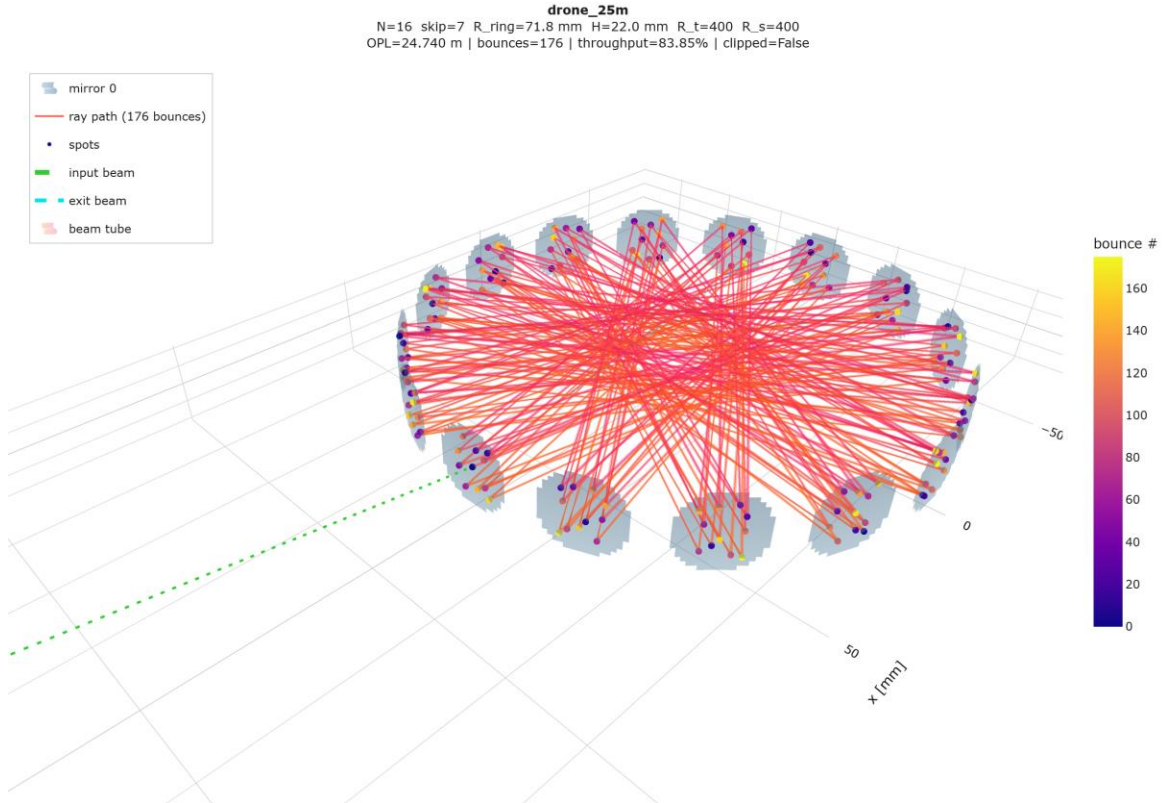


Fig. 3. The max-OPL corner (drone_25m): the same 16-mirror, 180 mm architecture driven to 176 chords (24.77 m, 11 spots per mirror). At $R = 0.999$ the 32 additional reflections relative to the headline design cost only 2.8 % of signal, which is why the long-path corner is preferred when the coating allows it.

4.2 The headline design

The 20.38 m design uses sixteen CM254-150 mirrors ($R = 300$ mm) on a ring machined to $R_{\text{ring}} = 72.155$ mm (envelope 180 mm, inner height 16 mm, sampled volume 0.26 L). The beam enters at 11.25° mean incidence, completes 144 chords of mean 141.5 mm, deposits 9 spots on each mirror with a worst-pair separation of 0.99 mm against summed beam radii of ~ 0.64 mm, keeps 4.9 mm of aperture margin, and re-crosses the hole plane within 6 nm of the entrance point at the designed bounce; all eight intermediate visits to M0 clear the hole by at least 1.46 mm beyond the local beam radius. Mode matching sets a 0.326 mm waist 102 mm past the hole (radius at the hole 0.365 mm; hole transmission 100.0 %). Transmission is $R^{143} = 86.7\%$ at $R = 0.999$: a 10 mW DFB source delivers ≈ 8.7 mW to the detector, so the photon budget is not the limiting factor at any realistic coating.

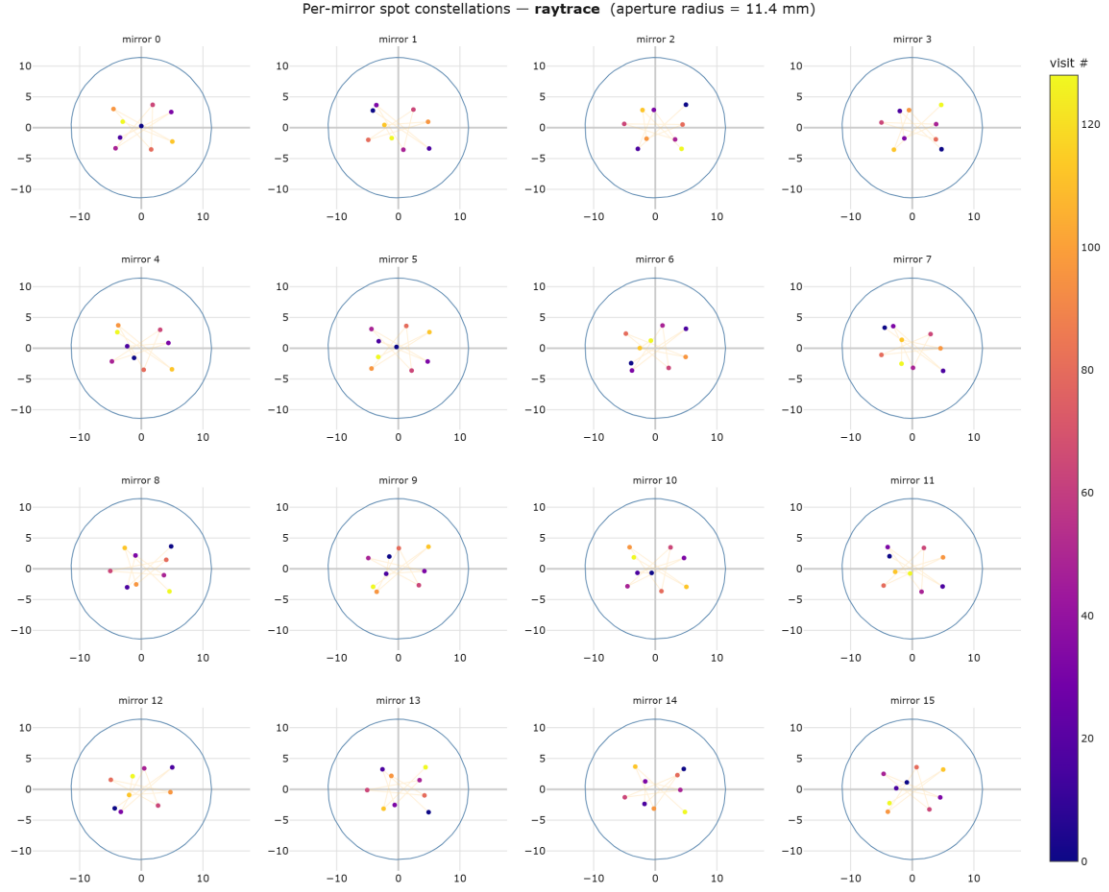


Fig. 4. Per-mirror spot constellations of the headline design (ray-traced): each of the 16 mirrors carries the same 9-point pattern at a different phase origin; the worst pair over all mirrors is the verified fringe-safety quantity. M0 additionally shows the entrance/exit hole.

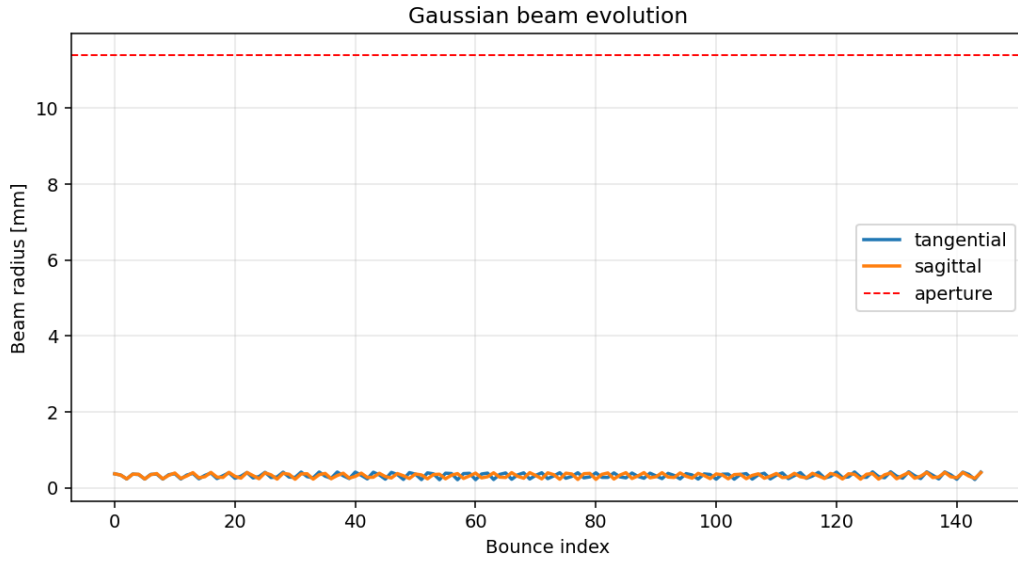


Fig. 5. Tangential and sagittal $1/e^2$ beam radii at each bounce under mode-matched injection: the beam rides the cell eigenmode at 0.21–0.42 mm for all 144 chords, an order of magnitude below the 11.4 mm clear aperture.

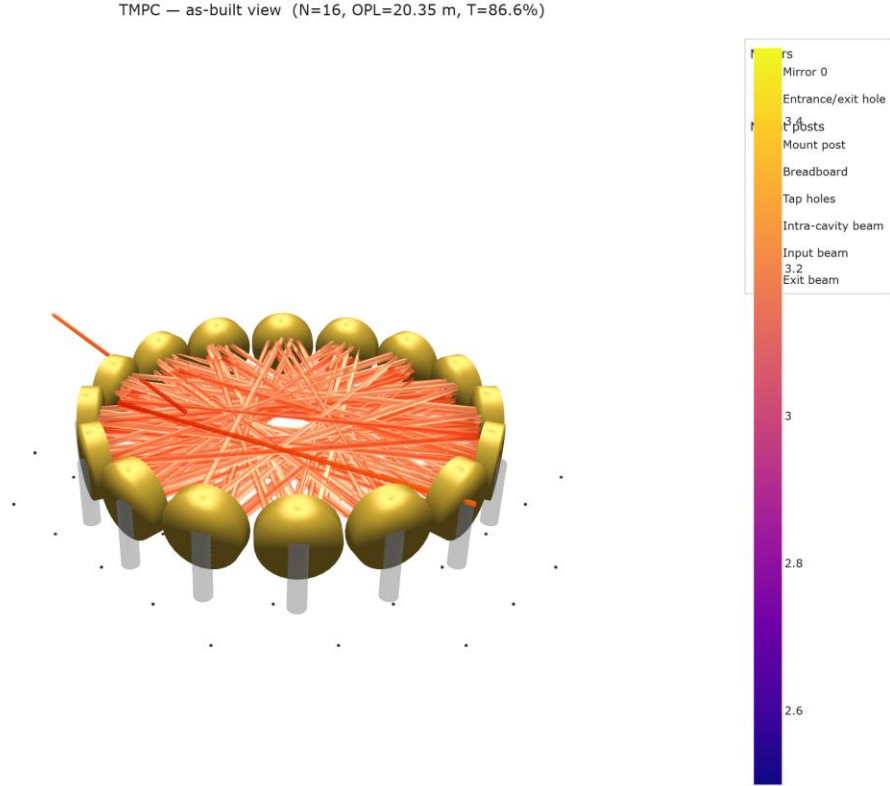


Fig. 6. As-built rendering of the headline design: sixteen 25.4 mm mirrors in a one-piece machined ring; laser collimator, mode-matching lens and detector all mount behind M0, the exit beam leaving 22.5° from the injection axis.

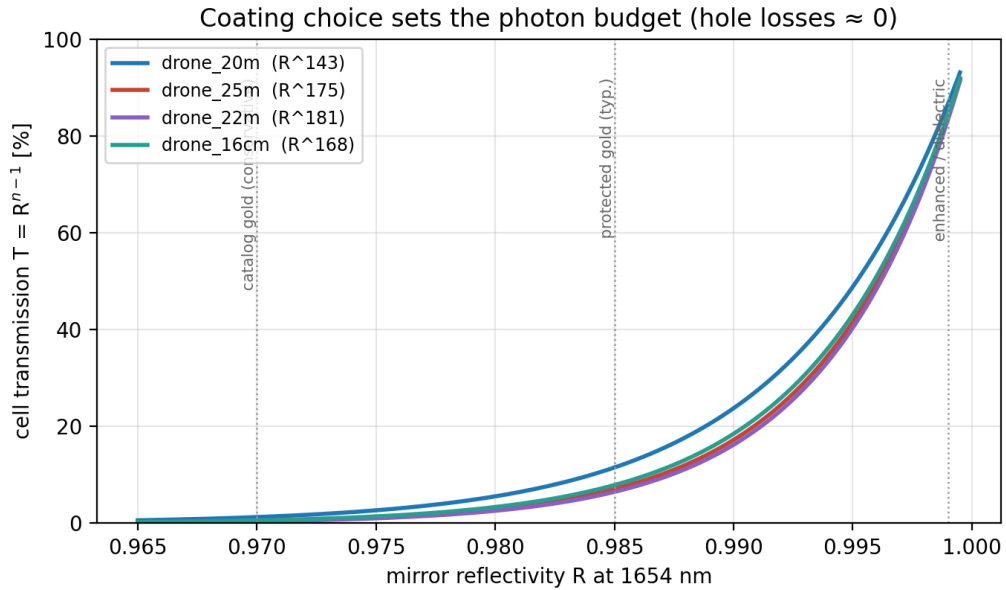


Fig. 7. Coating choice sets the photon budget: with the coupling hole rendered lossless by mode matching, transmission is exactly R to the power (chords $- 1$) for every design. Vertical guides mark conservative catalogue gold, typical protected gold, and enhanced/dielectric coatings.

4.3 Drone integration: mass and power budget

Table 2 gives a CAD-level mass estimate for the complete airborne sensor head. Mirror substrates are 25.4 mm $\times$ 6.35 mm glass ($\approx$ 8 g each); the one-piece aluminium ring (14 mm radial section, pocketed) and two 1.5 mm lids dominate the optical-head mass and admit $\sim$ 30 % lightweighting by skeletonising; injection optics comprise the fibre collimator, mode-matching lens and InGaAs photodiode on a common plate behind M0. With a compact DFB driver/DAQ stack the full payload is $\approx$ 1.0–1.2 kg, within the payload class of standard commercial multirotor platforms, and the 0.26 L sampled volume exchanges in well under a second at modest flow. Power is dominated by the laser TEC and electronics at a few watts; the cell itself is passive.

Table 2. Mass budget (CAD-level estimates, headline vs compact design).

item	drone_20m (Ø180)	drone_16cm (Ø159)
mirror substrates	16 $\times$ 8 g = 128 g	13 $\times$ 8 g = 104 g
Al ring (pocketed) + lids	$\approx$ 450–600 g	$\approx$ 380–500 g
injection optics + detector plate	$\approx$ 80 g	$\approx$ 80 g
optical head subtotal	$\approx$ 0.66–0.81 kg	$\approx$ 0.56–0.68 kg
DFB laser + driver + DAQ	$\approx$ 0.30–0.40 kg	$\approx$ 0.30–0.40 kg
total payload	$\approx$ 1.0–1.2 kg	$\approx$ 0.9–1.1 kg

5. Tolerance and engineering analysis

Each design was subjected to 300–400 Monte-Carlo trials with independent per-mirror perturbations at research-grade magnitudes (tilt σ = 0.5 mrad per axis; decentre σ = 50 μ m lateral, 100 μ m axial; curvature σ = 1 mm; ring radius σ = 0.1 mm; launch σ = 50 μ m and 0.5 mrad). No trial in any design clipped or lost the beam, and all completed the full traversal count; geometric quantities in Table 3 are coating-independent. One-at-a-time sensitivity identifies the dominant error source per design: mirror-pocket tilt for the $R \geq 300$ mm designs (3.3 mrad of exit drift per 0.5 mrad tilt σ , curvature error negligible), and ring radius / curvature for the compact $R = 100$ mm design (tilt negligible). The one-piece machined ring, whose pocket angles are cut in a single CNC operation, targets precisely the dominant term; the compact design additionally wants $\sim 2\times$ tighter machining or use of its thermal trim at first alignment.

Table 3. Monte-Carlo results at research-grade tolerances (mean $\pm$ σ ; p95 in brackets).

metric	drone_20m	drone_25m	drone_16cm
trials clipping / losing beam	0/400	0/300	0/300
OPL [m]	20.35 $\pm$ 0.03	24.74 $\pm$ 0.03	20.61 $\pm$ 0.03
exit pointing drift [mrad]	4.2 $\pm$ 2.6 (9.4)	5.0 $\pm$ 3.3 (11.2)	38 $\pm$ 23 (89)
spot-pattern walk p95 [mm]	0.89	1.43	2.95
hole clearance available [mm]	1.46	3.40	1.65

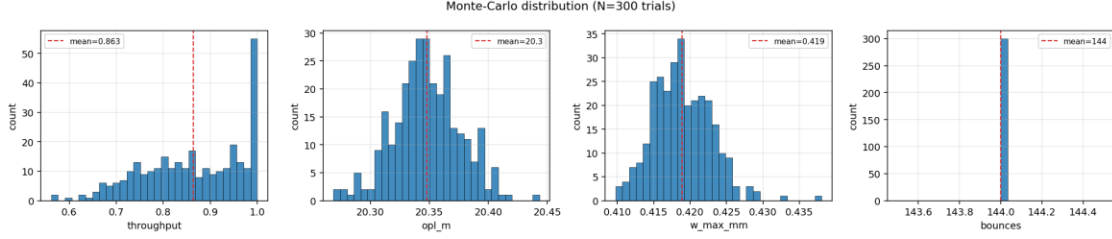


Fig. 8. Monte-Carlo distributions for the headline design at research-grade tolerances (400 trials): traversal count is invariant at 144; OPL and exit drift remain within operational bounds in every trial.

Systematic errors are compensated rather than tolerated. Catalogue mirrors carry a ± 1 % focal-length tolerance, typically biased across a delivered lot; re-optimising only the ring radius restores the complete check matrix across the entire band with a strictly linear trim — 0.72 mm per percent of curvature error for the 180 mm designs, 0.62 mm for the compact one. The assembly rule is therefore: measure the delivered curvatures, machine the ring to the interpolated radius, walk in the final micrometres with the temperature trim. Temperature acts through the same mechanism (aluminium scales R_{ring} by 23.6 ppm/K against an essentially fixed glass curvature): with the launch frozen, the headline design passes every check over ± 26 K on plain aluminium (± 20 K for the 24.8 m design, ± 8 K for the compact one), an invar ring holds $\geq \pm 30$ K for all three, and a small ring heater doubles as the closure fine-adjustment actuator. All designs also pass every check up to $M^2 = 1.3$, beyond typical fibre-coupled DFB sources.

Table 4. Systematic-error compensation and robustness.

design	ROC ± 1 % $\rightarrow$ ring trim	thermal window (Al)	thermal window (invar)	M^2 tested
drone_20m	± 0.72 mm (linear)	± 26 K	$\geq \pm 30$ K	1.3 pass
drone_25m	± 0.72 mm (linear)	± 20 K	$\geq \pm 30$ K	1.3 pass
drone_16cm	± 0.62 mm (linear)	± 8 K	$\geq \pm 30$ K	1.3 pass

6. Discussion

Table 5. Position against published compact multipass cells (demonstrated values unless noted).

cell	diameter	OPL	optics	notes
Tuzson 2013 [1]	~ 100 mm	4.1 m	monolithic diamond-turned	first monolithic toroidal
Graf 2018 [4]	< 200 g	10 m	segmented diamond-turned	interference-free
Chang 2020 [7]	100 mm	10 m (30 m theory)	ring surface	multi-layer patterns
Dong 2016 [9]	170 $\times$ 65 $\times$ 55 mm	54.6 m	2 custom spherical	densest; critical alignment
this work, compact	159 mm	20.6 m	13 $\times$ catalogue \$74	84.5 % T @0.999
this work, headline	180 mm	20.4 m	16 $\times$ catalogue \$74	86.7 % T @0.999
this work, max OPL	180 mm	24.8 m	16 $\times$ catalogue \$74	83.9 % T @0.999

Within the toroidal class the verified paths roughly double the demonstrated state of the art at comparable diameter, using catalogue optics and a single machined part; the astigmatic-Herriott record [9] remains denser but at custom-optics and alignment cost the one-piece ring eliminates. Three limitations bound the claims: the results are simulation-verified rather than experimentally demonstrated (the verification chain

— exact tracing, independent cross-validation, Monte-Carlo tolerancing, systematic-error compensation — is designed to de-risk the build, and machining-level specification sheets accompany each design); coating reflectivity enters as an external parameter, quoted parametrically throughout; and surface-figure error and scatter are not yet modelled, meriting measurement on the assembled cell.

7. Conclusion

Chord-skip ring geometry, engineered dual-plane re-entrance tuned by the machined ring radius, an exact constellation criterion, and mode-matched single-hole coupling together turn catalogue 1-inch mirrors into verified 20–25 m multipass cells inside drone-compatible 143–180 mm envelopes, transmitting ~84–88 % at $R = 0.999$ on a ≈ 1 kg payload. Next steps: fabricate the headline design, measure transmission and fringe floor against the simulated budgets, and integrate a 1654 nm DFB source for airborne methane sensing.

Data and code availability

Simulation platform, search harness, all design specifications, and scripts reproducing every figure, table and this document: github.com/Sumant-varanasi/toroidal-cell (_CURRENT/, presets drone_20m, drone_25m, drone_16cm). Acknowledgements and author list to be completed; per the project brief, intellectual property resides with Aston University.

References

- [1] B. Tuzson, M. Mangold, H. Looser, A. Manninen, and L. Emmenegger, “Compact multipass optical cell for laser spectroscopy,” *Opt. Lett.* 38(3), 257–259 (2013).
- [2] A. Manninen, B. Tuzson, H. Looser, Y. Bonetti, and L. Emmenegger, “Versatile multipass cell for laser spectroscopic trace gas analysis,” *Appl. Phys. B* 109, 461–466 (2012).
- [3] M. Mangold, B. Tuzson, M. Hundt, J. Jágerská, H. Looser, and L. Emmenegger, “Circular paraboloid reflection cell for laser spectroscopic trace gas analysis,” *J. Opt. Soc. Am. A* 33(5), 913–919 (2016).
- [4] M. Graf, L. Emmenegger, and B. Tuzson, “Compact, circular, and optically stable multipass cell for mobile laser absorption spectroscopy,” *Opt. Lett.* 43(11), 2434–2437 (2018).
- [5] M. Graf et al., coherent-field modelling and absorption-mask fringe suppression in toroidal multipass cells (2017). [verify]
- [6] K. Sun et al., double-row circular multipass cell reaching 74.7 m (2018). [verify]
- [7] J. Chang et al., “Multi-layer beam patterns from a single toroidal ring surface,” *Opt. Lett.* 45 (2020). [verify]
- [8] Y. Kilinc et al. (CREMA collaboration), cryogenic toroidal multipass cavities at 6.8 μm with >15 m path (2026). [verify]
- [9] L. Dong et al., ultra-compact dense-spot-pattern multipass cell, 54.6 m in 220 mL (2016). [verify]
- [10] K. Liu et al., dense-pattern multipass cell, 26.4 m (2015). [verify]
- [11] D. R. Herriott, H. Kogelnik, and R. Kompfner, “Off-axis paths in spherical mirror interferometers,” *Appl. Opt.* 3(4), 523–526 (1964).
- [12] D. R. Herriott and H. J. Schulte, “Folded optical delay lines,” *Appl. Opt.* 4(8), 883–889 (1965).
- [13] J. B. McManus, P. L. Keabian, and M. S. Zahniser, “Astigmatic mirror multipass absorption cells for long-path-length spectroscopy,” *Appl. Opt.* 34(18), 3336–3348 (1995).
- [14] Thorlabs Inc., CM254-xxx-M01 concave mirror series, product datasheets (2026).
- [15] H. Kramer, Optiland: open-source sequential ray tracing in Python, github.com/HarrisonKramer/optiland.